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Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Use substitution to find the Taylor series at $x = 0$ of the function $6 \sin(-x)$.

Compare the given function to $\sin(u(x))$. Identify $u(x)$.

$$6 \sin(u(x)) = 6 \sin(-x) \\ u(x) = -x$$

Let $u = -x$. To find the Taylor series of $6 \sin(-x)$ at $x = 0$, substitute $x = 0$ into the expression for u and simplify.

$$u = -(0) \\ = 0$$

Therefore, if $u = -x$, then the Taylor series of $6 \sin(-x)$ at $x = 0$ is the Taylor series of $6 \sin u$ at $u = 0$. Find the Taylor series of $6 \sin u$ at $u = 0$, in terms of u (not x).

$$\sum_{n=0}^{\infty} \frac{6(-1)^n (u)^{2n+1}}{(2n+1)!}$$

Substitute $u = -x$ into the general expression for the n th term in the Taylor series below.

$$\sum_{n=0}^{\infty} \frac{6(-1)^n (u)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{6(-1)^n (-x)^{2n+1}}{(2n+1)!}$$

Therefore, the general expression for the n th term in the Taylor series at $x = 0$ for $6 \sin(-x)$ is shown below.

$$\sum_{n=0}^{\infty} \frac{6(-1)^n (-x)^{2n+1}}{(2n+1)!}$$